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PREDICTING GOLD PRICE IN INDIA USING SARIMA MODEL

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ABSTRACT

The time series approach has been considered into account for predicting Gold Price in India. The secondary data is taken from the **indiastat** website from the year 1986 to the year 2018 for the period of 33 years. Error measures such as Root Mean Square Error, Mean Absolute Percentage Error, and Mean Absolute Error are used to validate the model. SARIMA model (0,1,0) $(1,0,1)^{12}$ finalized as the best fit for the taken dataset. The study helps the decision makers, investors and commodity traders to take the prior proclamation.

Keywords: RMSE, MAE, MAPE, Seasonal Difference, SARIMA Model.

I. INTRODUCTION

Gold is regarded as one of the rarest, most beautiful, and most useful metals in history. Its use may be traced back thousands of years to the time of ancient civilizations including the Greeks, Romans, and Egyptians. An overview of this intriguing metal may be found here, Gold has a brilliant, golden hue and is a thick, soft, and malleable metal. It is a great conductor of heat and electricity because it is very ductile, or able to be drawn out into thin wires. "Au," which stands for gold in the periodic table, is derived from the Latin word "aurum," which means "shining dawn". At 0.004 parts per million, gold is comparatively uncommon in the Earth's crust. It is mainly found in alluvial deposits or subterranean veins. For thousands of years, gold mining has been a prominent human activity. Techniques used range from conventional panning to large-scale industrial operations. Throughout history, gold has been deeply significant to many different civilizations. It has been used as money, in royal regalia, and in religious events. Due of its rarity and timeless beauty, it has come to represent prestige, power, and prosperity.

The studies related to the present study in the past ages are revealed below, Chaku et al., (2022) was applied SARIMA model and forecast average monthly gold prices. Information about gold was acquired from January 2015 to December 2020. For the analysis, monthly adjusted close prices were employed. Data on gold prices remained unchanged following the initial difference. It was determined that the best model to match the gold price data with the lowest AIC and BIC was SARIMA (0,0,0)(0,1,1)[12].A forecast was obtained for the price of gold from January 2021 to December 2025.Over the projection period (2021–2025), the average monthly gold price is expected to climb and fall. The January through June 2021 actual values were compared to the forecast values. The real gold prices and the expected values did not differ significantly. Prospective investors ought to think about gold in their portfolios as a diversifier, a store of value, and as a hedge against the price fluctuations this analysis predicts.

Makala et al., (2021) used SVM and ARIMA models in this article's research for prediction. The analysis of the report is based on daily data from the World Gold Council spanning 1979 to 2019. Models are trained using data up to 2014; the remaining data are used for validation. Using the performance assessment tools of RMSE and MAPE, the study's findings demonstrate that the SVM outperforms ARIMA, with RMSE of 0.028 and MAPE of 2.5 for the SVM and 36.18 and 2897 for ARIMA, respectively. The results indicate that, because of his great accuracy, SVM should be used to estimate the price of any commodity.

Tripathy et al., (2017) used the ARIMA (Auto Regressive Integrated Moving Average) model to anticipate the price of gold in India during a 25-year period, from July 1990 to February 2015.The accuracy of the model is also assessed by the study using Mean Absolute Error, Root Mean Square Error (RMSE), Maximum Absolute Percentage Error, Maximum Absolute Error (Max AE), and Mean Absolute Percentage Error (MAPE). The study's findings indicate that, because ARIMA (0, 1, 1) has the least amounts of MAPE, Max AE, and MAE, it is the best model to utilize when predicting Indian gold prices. According to the study, the price of gold one month ago has a big influence on the price of gold today. The study's findings are especially significant to economists,

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market regulators, policy makers, and investors because they shed light on how well the price of gold performs, which helps them make better investment decisions and create better risk management strategies.

Guha et al., (2016) provides an inside look at the implementation of the ARIMA time series model to anticipate the future gold price in Indian browsers based on historical data from November 2003 to January 2014. Thus, the purpose of this advice is to advise investors on when to purchase and sell yellow metal. This financial instrument has become increasingly popular recently as a result of the Indian economy being impacted by factors such as shifting political landscapes, global cues, high inflation, and so on. As a result, researchers, investors, and speculators are looking for alternative financial instruments to diversify their portfolios and reduce risk. In the past, gold was only bought in India during weddings or other special occasions. However, as gold has grown in value in the eyes of investors, it is now vital to forecast gold prices using an appropriate methodology.

II. MATERIALS & METHODOLOGY

Data Source

The source of the data is gathered from the **indiastat** website. There are 396 monthly gold price observations of 10 grams in India between 1986 and 2018 for a period of 33 years.

Seasonal Autoregressive Integrated Moving Average Model (SARIMA)

The ARIMA model is for non-seasonal non-stationary data. Box and Jenkins have generalized this model to deal with seasonality. Their proposed model is known as the Seasonal ARIMA (SARIMA) model. In this model seasonal differencing of appropriate order is used to remove non-stationarity from the series. A first order seasonal difference is the difference between an observation and the corresponding observation from the previous year and is calculated,

$$Z_t = Y_t - Y_{t-s}$$

For monthly time series $s = 12$ and for quarterly time series $s = 4$, this model is generally termed as the SARIMA $(p, q, d) * (P, Q, D)$ model. Here t, s is the seasonally differenced series.

III. RESULTS & DISCUSSION

The time series analysis is carried out for given data to forecast gold price for future periods of the time series data against the various time periods. Identify the appropriate seasonal ARIMA model for the data and estimate the parameters for time series of monthly exports of gold price in India. After that, perform diagnostic checks of the residuals for autocorrelation, variance and normality assumptions. Finally, the appropriate seasonal ARIMA model for forecasting can be used.

The behaviour of the data is in upward trend over time. The time series plot of monthly data of gold price in India is displayed in figure 1 below,

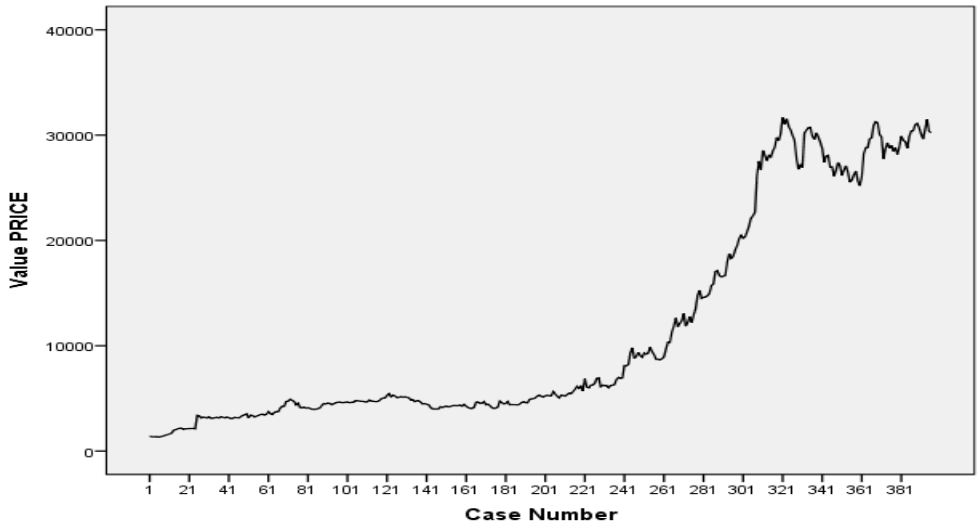

Figure 1. Time Series plot of Gold Price in India (Trend)

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Figure 1 indicates that the time series is not stationary series. The plot shows consist pattern of short term changes for data which indices the existence of seasonal fluctuations. This series varies randomly over time then there is a trend and seasonal fluctuations. Gold price of the 396 monthly data, the price value is having upward trend in the observed from the time series data. Investigation is done by examination of autocorrelation and partial autocorrelation function as shown in below figure 2.

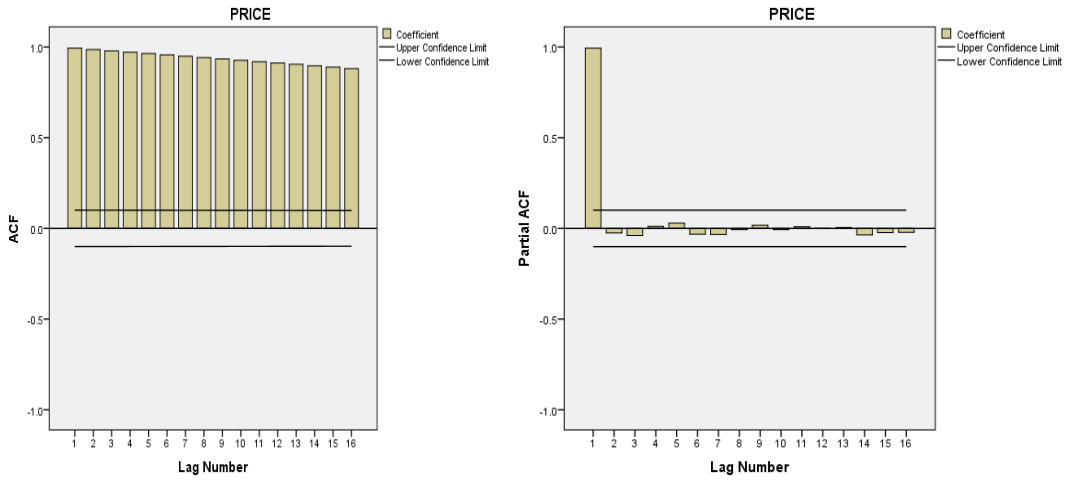

Figure 2. ACF & PACF plots of dataset

All the above results , displays autocorrelation plot and Partial ACF plot of the data against lag t which shows that the first several autocorrelation coefficients are persistently large and trailed off to zero rather slowly and indicates the time series data in non-stationary and does not vary about a fixed level. Hence, it is necessary to transform the non-stationary into stationary. Taking first difference, is for eliminating the trend and creating a stationary mode.

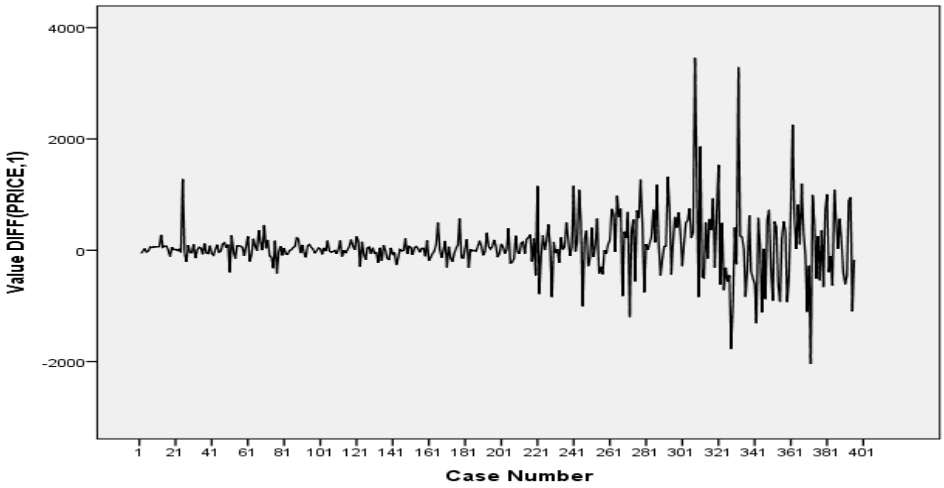

Figure 3. Time Series plot for first order differencing

This first order difference of gold price rate for the year 1986 to 2018, gold price value is having a stationary trend in the observed from time series, to find the seasonal variation of this data of graph is given below,

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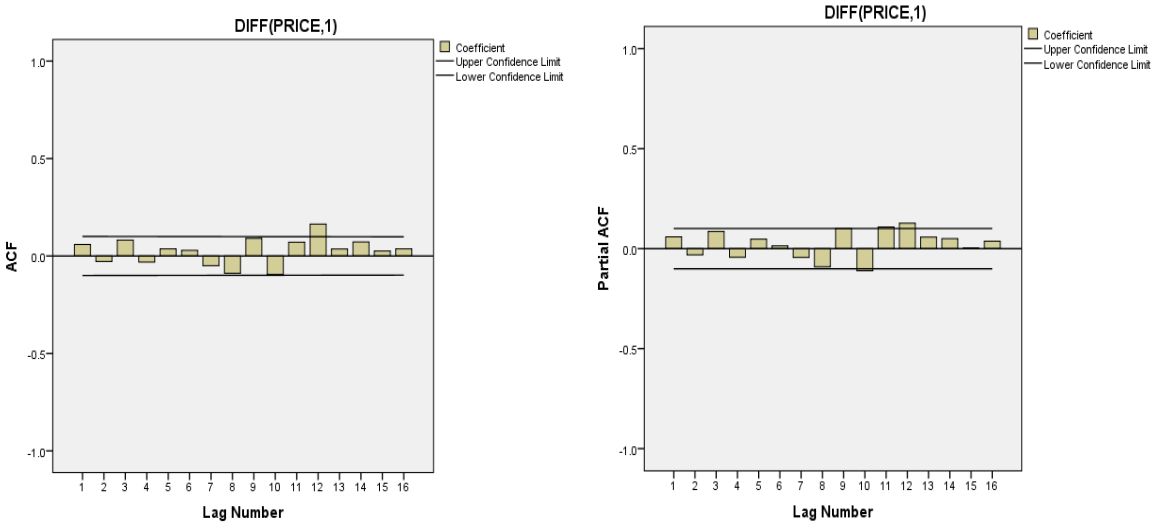

Figure 4. ACF & PACF for first order difference

To remove the effect of seasonal component, the seasonal difference is found in below figure 5.

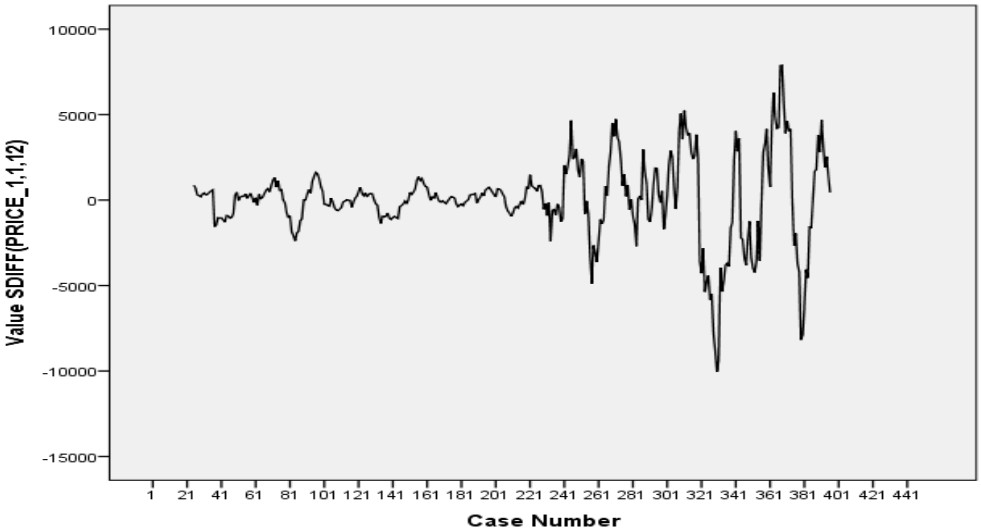

Figure 5. Time Series plot for seasonal difference

Hence, from the above plot, it is confirmed that the have been transformed into stationary data. Then find autocorrelation and partial autocorrelation for transformed data.

Model Identification

It shows how determine the order of seasonal ARIMA model and identify the model specifications such as ACF and PACF in the addition to BIC, MSE, RMSE, MAPE and MAE criteria. It can be seen from the autocorrelation coefficients and partial autocorrelation coefficient of the series that it is necessary to consider the seasonal changes when identifying and estimating the model.

Parameter Estimation

Since it is concluded that the SARIMA $(0,1,0)(1,0,1)^{12}$ model is the best model, the parameter has been estimated using the method of maximum likelihood estimation as it is the best and most approximate model of estimation. The result of the parameter estimation of the model is shown in the table 1.

Table 1. Parameter Estimates of ARIMA model

Variable	Estimate	SE	T	Sig.
Constant	0.008	0.004	2.165	0.031
MA lag 1	0.358	0.05	6.18	0.000

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AR, Seasonal Lag 1	0.822	0.085	9.686	0.000
MA, Seasonal Lag 1	0.671	0.113	5.933	0.000

Table 2. Error Measures of ARIMA model

Criteria	RMSE	MAPE	MAE	BIC	R^2
SARIMA $(0,1,0)(1,0,1)^{12}$	516.860	2.858	307.216	12.541	0.997

It shows in table 2, that the p-value for the coefficient are less than α=0.05. This indicates that the coefficient is significantly different from zero. Therefore, this supports that this model is appropriate.

Thus, the final model is SARIMA $(0,1,0)(1,0,1)^{12}$ can be expressed as,

$$(1-0.822B\text{^}12)(1-B\text{^}12)(1-B)\,rt \;=\; (1-0.358B)\,(1-0.671B\text{^}12)\,\varepsilon t$$

Figure 6, shows the estimates of ACF & PACF for the residual of this model. It indicates that the parameters of the ACF & PACF are all between the upper and lower bound of confidence interval.

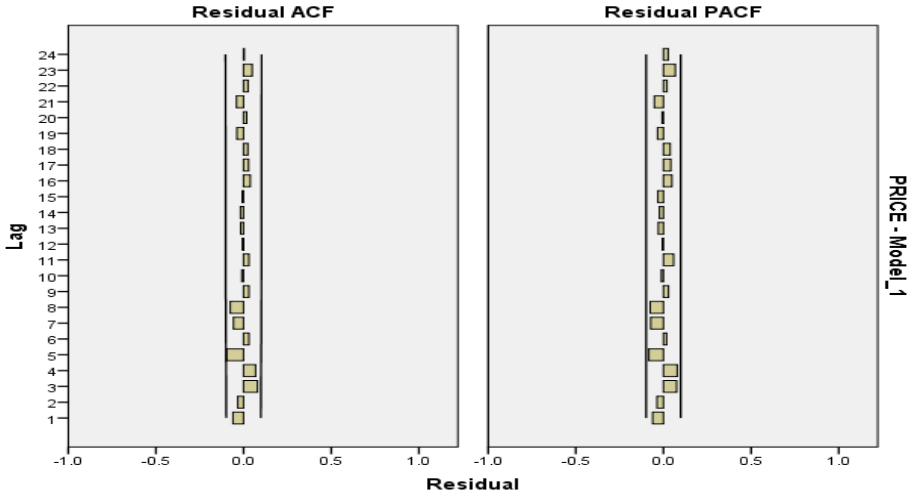

Figure 6. ACF & PACF Residuals of the fitted model

Forecasting

The final model for the monthly gold price in India than has been expressed by using SARIMA model. This research used to forecast the future gold price. The forecasting of time series for monthly price of gold in India has been plotted below figure 7.

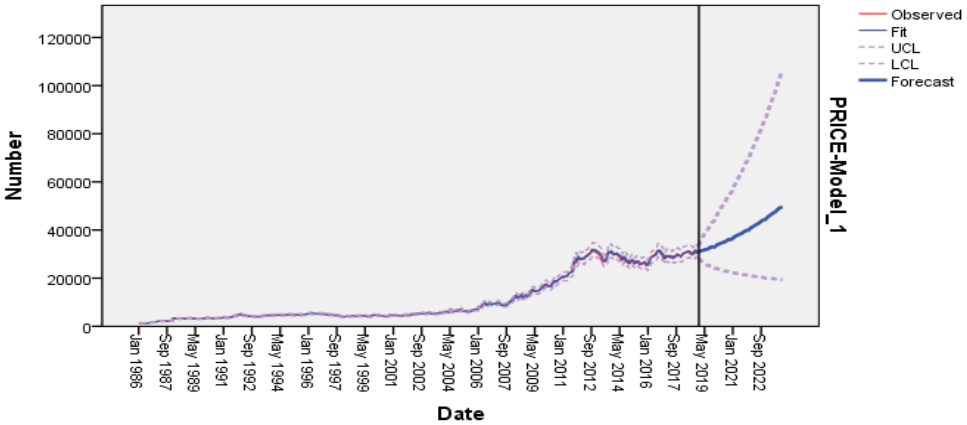

Figure 7. Forecasted Graph of the dataset

Figure 7, shows the result that the series of the forecasted value follow the same behaviour of the original series of monthly export of gold price in India. The results of the forecasted values is shown in the below table (3) for the years 2019-2023 are all between the upper and lower boundaries of the 5% significance of limits. This confirms that the forecasting is very efficient.

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Table 3. Forecasted future values for the years 2019-2023

Years	Month	UCL	LCL	Forecasted values (10 grams)
2019	January	33509	28172	30755
	February	35269	27596	31258
	March	36337	26907	31360
	April	37472	26487	31627
	May	38462	26096	31836
	June	39276	25680	31945
	July	39975	25262	31996
	August	41016	25112	32344
	September	42212	25086	32827
	October	43068	24884	33057
	November	43513	24477	32987
	December	43882	24061	32876
2020	January	45276	24036	33419
	February	46628	24003	33935
	March	47477	23732	34093
	April	48521	23579	34399
	May	49484	23403	34653
	June	50311	23179	34819
	July	51054	22933	34933
	August	52182	22871	35315
	September	53492	22893	35819
	October	54464	22774	36096
	November	55024	22494	36104
	December	55517	22201	36074
2021	January	58533	22221	36637
	February	59520	22038	37176
	March	60719	21941	37392
	April	61845	21822	37744
	May	62841	21663	38049
	June	63759	21485	38275
	July	65080	21446	38455
	August	66593	21470	38878
	September	67767	21385	39412
	October	68525	21174	39742
	November	69220	20950	39828
	December	70955	20972	39882

2022	January	72664	20984	40474
	February	73862	20849	41046
	March	75284	20779	41326
	April	76638	20691	41729
	May	77864	20572	42091
	June	79017	20436	42383
	July	80586	20409	42632
	August	82356	20430	43105
	September	83785	20365	43679
	October	84797	20201	44068
	November	85750	20028	44237
	December	87752	20049	44375
2023	January	87752	19960	45008
	February	89732	19908	45623
	March	91200	19842	45972
	April	92900	19750	46435
	May	94535	19645	46862
	June	96048	19626	47224
	July	97490	19645	47550
	August	99362	19596	48081
	September	99441	19468	48706
	October	103180	19332	49163
	November	104502	19459	49418
	December	104769	19960	49645

Therefore, it is concluded from the table 3, there is an increasing trend in the gold price in India from the year 2019 to the year 2023.

IV. CONCLUSION

In this study, a univariate time series data on monthly standard gold price (10grams) of the Mumbai market have been taken from Jan 1986 to Dec 2018. The data have been examined and various SARIMA model combinations were tried fit to the dataset and concluded that the SARIMA (0,1,0) $(1,0,1)^{12}$ is the best fit for the dataset. This article can be useful for commercial traders and gold investors to take prior decisions.

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